

NATIONAL COLLEGE OF IRELAND — SCHOOL OF COMPUTING

Master of Science in Data Analytics (MSCDAD) | Research Practicum WEEKLY PROGRESS REPORT — WEEK 1

Student Name:	Rohitkumar Amritlal Jaiswal (ID: x25119613)
Supervisor / Lecturer:	Dr. Thanos Staikopoulos
Project Title:	Coupling Port Congestion & Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation
Reporting Period / Date:	Week 1 (Kickoff, Technical Initialization & Baseline Benchmark) Tuesday, 29 September 2026

1. Activities Completed This Week (What?)

- **Project Proposal & Ethics Sign-Off:** Finalised the 4,691-word Research in Computing Proposal PDF (CA2) streamlining scope into 3 coupled modules (C1, C2, C3). Curated 7 peer-reviewed reference papers and signed the official Ethics Consideration Form.
- **Repository Architecture & Packaging:** Initialized the formal Git repository with modern PEP 621 packaging (irishlogix v0.1.0) and automated testing harness.
- **Automated CSO Data Pipeline (cso_loader.py):** Integrated the Central Statistics Office (CSO) PxStat API; successfully ingested and parsed 444 vessel arrivals (TBQ01), 2,331 Ro-Ro freight records (TBQ04), and 3,600 trade region records (TBQ05). Created unified port congestion matrix.
- **Irish Freight Corridor Graph (corridor_network.py):** Implemented NetworkX road network graph with 11 strategic nodes (Dublin, Cork, Rosslare, M50, Athlone, Limerick, Galway, Waterford, Dundalk) and 15 bidirectional corridors with tolls, speed limits, and distance matrices.
- **Baseline FIFO Dispatcher Simulator (baseline_dispatcher.py):** Coded and executed empirical baseline simulation across 50 benchmark consignments: 8,636.5 km distance, 104.4 driving hrs, **63.5 idle waiting hrs** (averaging **1.27 hrs wasted idling per trip**), 2,879.3 L diesel, 7,601.3 kg CO2e, and €14,791.21 total cost (avg €295.82/trip).
- **Unit Test Suite:** Authored 9 unit tests across data ingestion, graph routing, and simulation; 100% passing (pytest tests -v in 1.08s).

2. Evaluation & Critical Reflection (So What?)

- **Validation of Research Gap:** The Week 1 baseline simulation empirically validates the core research rationale: without predictive delay awareness, freight hauliers waste an average of 1.27 hours per shipment waiting at terminal gates and customs inspections, creating direct financial and carbon penalties.
- **Data Grounding:** Live CSO PxStat API data guarantees verified, open historical maritime records (2017–2026) for Dublin, Cork, and Rosslare, de-risking Component 1 model development.

3. Action Plan & Next Steps (Now What?)

- **Oral Pitch (1 October):** Deliver the timed 1–2 minute project pitch adhering to Problem → Gap → Idea → Method → Contribution.
- **Supervisory Alignment:** Present Week 1 code and benchmark findings to Dr. Thanos Staikopoulos during scheduled supervisory review.
- **EMODnet & AIS Ingestion:** Incorporate EMODnet vessel density metrics into the maritime congestion feature store.

4. Supervisor Meeting Log & Agreed Actions

Agenda Items:	1. CA2 Proposal & Ethics sign-off. 2. 1 October Pitch rehearsal. 3. Demonstration of Week 1 technical deliverables: Git repo, CSO pipeline, baseline FIFO simulation.
Supervisor Guidance:	• Emphasize empirical gap between unbuffered spreadsheet planning and predictive optimization. • Ensure baseline simulation parameters are grounded in Irish logistics economics.
Agreed Actions:	1. Deliver 1–2 minute pitch on 1 October. 2. Submit weekly progress updates every Tuesday. 3. Begin feature engineering on ingested CSO port traffic data.

5. Project Milestone & Gantt Tracking

M1	Week 3	ADVANCED / ON SCHEDULE	Baseline FIFO model operational; CSO data ingested; corridor graph built
M2	Week 9	PLANNED	XGBoost models (C1, C2) coupled with Pareto Optimizer (C3)

M3	Week 13	PLANNED	Dual-system comparative evaluation & Streamlit UI dashboard
M4	Week 15	PLANNED	Final MSc Dissertation, code repository, and Viva Voce

Student Signature: Rohitkumar Amritlal Jaiswal

Date: 29 September 2026

Supervisor Signature: _____

Date: _____